

AP STATISTICS

FORMULA & INFERENCE

DECISION GUIDE

Stop Memorizing. Master the AP Exam.

You do not need another dense 600-page prep book. Designed for high-yield clarity, this compa

Inside This Hardcover Edition:

- * Plain-English Formulas: Every parameter encoded without proofs.
- * Unit Mindmaps: Visual 1-page concept road maps for all 9 units.
- * Master Mnemonics: Zero-guesswork checklists (PANIC, PHANTOM, BINS).
- * FRQ Sentence Frames: Standardized phrases for all 9 units.
- * Top 25 Traps: Never confuse 'failing to reject' with 'accept'.
- * Exam Pacing Strategy: Pacing blueprints for the exam.
- * Free Digital Companion: Instant interactive mobile app.

AP STATISTICS FORMULA & INFERENCE DECISION GUIDE BOOK 1

STATISTICAL FORMULAS

$$\bar{x} = \frac{\sum x_i}{n}$$

$$s = \sqrt{\frac{\sum (x_i - \bar{x})^2}{n-1}}$$

$$z = \frac{\bar{x} - \mu}{\frac{s}{\sqrt{n}}}$$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

CONFIDENCE INTERVALS

$$\bar{x} \pm z^* \frac{\sigma}{\sqrt{n}}$$

$$\hat{p} \pm z^* \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

$$\bar{v} \pm z_0^* \frac{x - \mu}{\sigma}$$

Find Critical Values Like Key:

90% Confidence Level	$z^* = 90\% \rightarrow 1.0001$
95% Confidence Level	$z^* = 95\% \rightarrow 1.0093$
99% Confidence Level	$z^* = 99\% \rightarrow 1.0098$

SIGNIFICANCE TESTS

$p < 0.05$ $p < 0.05$

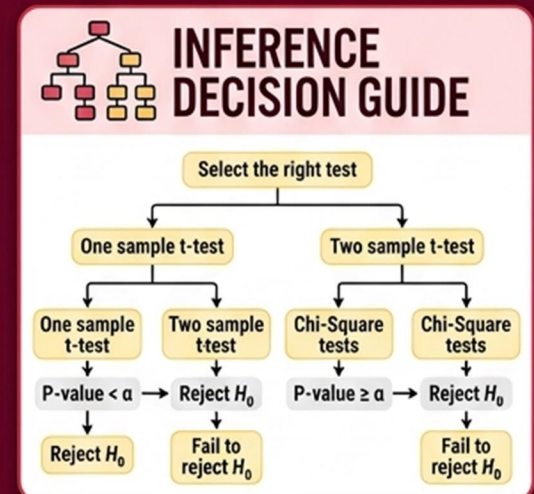
1. State: Hypotheses
2. Plan: Check Conditions
3. Do: Calculate Test Statistic & P-value
4. Conclude

t-test

$$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}$$

z-test statistic

$$z = \frac{\bar{x} - \mu_0}{\sigma}$$



[KDP BARCODE EXCLUSION ZONE]

CRACK THE EXAM WITH CLEAR FORMULAS, TABLES, AND STRATEGIES